How Ricardian Are We?

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   - Many endogenous regressors $\Longrightarrow$ many well-identified shocks

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   - **Not very!** Households internalize 9.2% of future taxes
   - GE model: government borrowing financed largely from abroad
   - Explains 71.0% of the decline in net foreign assets (NFA)

Theoretical Framework

"The debts of a nation are debts due from the right hand to the left."

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"The people who pay the taxes never so estimate them, and therefore do not manage their private affairs accordingly."

David Ricardo (1820)

Behavioral Consumption Function Ingredients

- Government budget constraint:

$$B_{t-1} + G_t = T_t + Q_t B_t$$

- Household budget constraint:

$$B_{t-1} + R_t^K K_{t-1} + Y_t^N = C_t + T_t + Q_t B_t + K_t$$

- Household Euler equation:

$$Q_t = \beta \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \right] + Z_t^d$$

$\tilde{\mathbb{E}}_t$: subjective expectations; Z_t^d: exogenous wedge

Behavioral Consumption Function Ingredients (Linearized)

- Government budget constraint:

$$b_{t-1} = \tau_t - g_t + \overline{B} q_t + \beta b_t$$

- Household budget constraint:

$$n_{t-1} + y_t = c_t + \tau_t + \overline{B} q_t + \beta n_t$$

- Household Euler equation:

$$q_t = \beta \tilde{\mathbb{E}}_t \left[\frac{\gamma}{\overline{C}} (c_t - c_{t+1}) \right] + z_t^d$$

Lowercase: level deviations; n_t: financial net worth; γ: relative risk aversion

Behavioral Consumption Function: Expectations

- What is $\tilde{\mathbb{E}}_t$?
  - Distorts the rational expectation $\mathbb{E}_t$
  - Can apply to all time series or a subset
  - Can allow naive or sophisticated expectations

- Rational and perceived present values:

$$v_t^\tau = \tau_t + \beta \mathbb{E}_t[v_{t+1}^\tau] \qquad \tilde{v}_t^\tau = \tau_t + \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]$$

- Perceived P.V. of future taxes proportional to the rational P.V.:

$$\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] = \theta \mathbb{E}_t[v_{t+1}^\tau]$$

θ: *behavioral attenuation* (key parameter)

The Behavioral Consumption Function

Proposition
Under the budget constraints and tax-forecast restriction,

$$c_t = (1-\beta)\left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1-\theta)\tau_t - \theta v_t^g + \theta \overline{B} v_t^q\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}_t^q + \zeta_t$$

- ζ_t: exogenous demand shock
- $\theta = 1$: Ricardian equivalence
- $\theta < 1$: consumption responds to debt-financed transfers

Empirical Strategy

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- Estimate the consumption function with time series data

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- Several challenges $\Longrightarrow$ develop **B-HIVE**

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- Missing data: macro shocks have mixed coverage
  - Especially problematic with many endogenous regressors
- Famously weak instruments
- Expectations appear in the structural equation
  - Long horizons in v_t^y, v_t^g, v_t^q

The Bayesian Hybrid Instrumental Variable Estimator



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B-HIVE: State-Space Components

1. A structural equation:

$$c_t = \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \phi_y y_t + \phi_q q_t + \sum_{j \in \{y,g,q\}} \phi_{vj} v_t^j + \zeta_t$$

$$\zeta_t = \rho_\zeta \zeta_{t-1} + h_c \varepsilon_{c,t}$$

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2. A VAR for $X_t = (n_t, b_t, \tau_t, y_t, g_t, q_t)'$:

$$X_t = \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G\varepsilon_t$$

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3. Instruments as noisy measurements of structural shocks:

$$w_t = \mu_w + M_X X_{t-1} + M\varepsilon_t + \eta_t$$

Instruments exclude the consumption innovation $\varepsilon_{c,t}$

Application

Data

- US quarterly data, 1951:Q4–2019:Q4
  - Consumption, personal income, net taxes, government spending
  - Household net worth, market value of federal debt, real T-bill rate
  - Nominal quantities relative to CBO nominal potential GDP
- 28 instruments from 19 sources, six shock categories
- Benchmark: rational present values + current income and interest rate
  - Freely estimated coefficients; four VAR lags
  - Alternatives: survey and yield-curve proxies

Structural Shock Instruments

External, High-frequency, Narrative 28 instruments

Monetary Policy (8): Jarociński and Karadi (2020); Miranda-Agrippino and Ricco (2021); Bauer and Swanson (2023); Swanson (2024), Aruoba and Drechsel (2024)

Government Spending (7): Fisher and Peters (2010), Ramey (2016); Romer and Romer (2016); Fieldhouse et al. (2018); Fieldhouse and Mertens (2023)

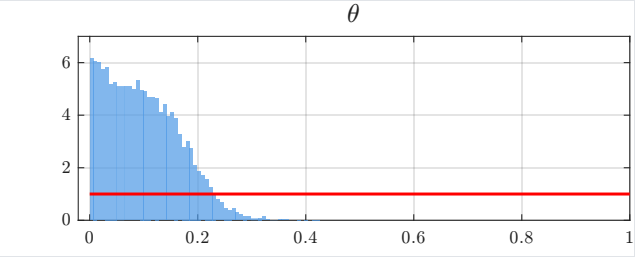
Tax/Borrowing (8): Leeper et al. (2012), Phillot (2025), Mertens and Ravn (2012); Lieb et al. (2024)

Technology (2): Fernald (2014); Miranda-Agrippino et al. (2025)

Oil (2): Kilian (2008), Känzig (2021)

Uncertainty (1): Piffer and Podstawski (2018)

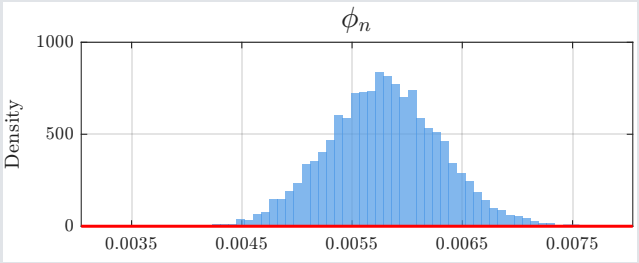
Posterior Distributions: Not Very Ricardian!



- θ: median 0.092
  - 90% interval: $[0.008, 0.218]$

Red: prior; blue: posterior. Point estimates are posterior medians.

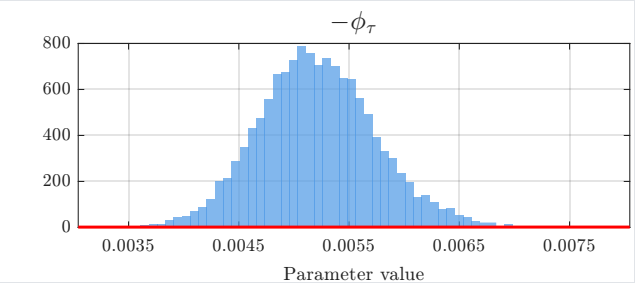
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- MPC-DFT: 0.519%
  - $-\phi_\tau$; consumption per dollar transferred

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Robustness: Alternative Consumption Functions

Specification	θ	Annual wealth MPC ($4\phi_n$)	MPC-DFT ($-\phi_\tau$)
Benchmark	0.092 [0.008, 0.218]	0.0232 [0.0198, 0.0265]	0.00519 [0.00436, 0.00614]
Imperfect risk sharing	0.101 [0.010, 0.368]	0.0154 [0.0115, 0.0202]	0.00343 [0.00205, 0.00463]
Distortionary taxes	0.335 [0.077, 0.674]	0.0159 [0.0123, 0.0198]	0.00266 [0.00112, 0.00408]
Nonseparable utility	0.263 [0.072, 0.447]	0.0217 [0.0184, 0.0256]	0.00397 [0.00320, 0.00481]
External habits	0.189 [0.040, 0.368]	0.0559 [0.0485, 0.0648]	0.01144 [0.00845, 0.01383]

Posterior medians; 90% credible intervals in brackets.

- Imperfect risk sharing: measured Berger–Bocola–Dovis wedge

Robustness: Alternative Implementation Choices

Specification	θ	Annual wealth MPC ($4\phi_n$)	MPC-DFT ($-\phi_\tau$)
Benchmark	0.092 [0.008, 0.218]	0.0232 [0.0198, 0.0265]	0.00519 [0.00436, 0.00614]
Six instruments	0.172 [0.015, 0.587]	0.0230 [0.0185, 0.0272]	0.00468 [0.00221, 0.00612]
Income-sum restriction	0.083 [0.006, 0.339]	0.0290 [0.0241, 0.0350]	0.00645 [0.00481, 0.00804]
Survey forecasts	0.562 [0.057, 0.954]	0.0154 [0.0113, 0.0219]	0.00169 [0.00017, 0.00413]
Larger X_t vector	0.076 [0.006, 0.258]	0.0230 [0.0191, 0.0269]	0.00520 [0.00399, 0.00633]

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- Six instruments: one high-quality structural shock per category
- Additional checks: alternative IVs, alternative detrending, similar conclusions

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  - Estimated annual wealth MPC: 2.32%

General Equilibrium

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  - How much can our estimate explain?

GE Model Summary

- Domestic economy: neoclassical production and capital accumulation

$$Y_t = K_{t-1}^{\alpha} A_t^{1-\alpha} \qquad K_t = (1-\delta)K_{t-1} + I_t$$

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  - Endogenous bond price Q_t

Government Debt and Expectations

- Government budget constraint:

$$G_t + B_{t-1} + B^*_{t-1} = T_t + Q_t(B_t + B^*_t)$$

- Debt relative to productivity follows an AR(1):

$$\frac{B_t + B^*_t}{A_t} - \overline{b} = \rho_B \left(\frac{B_{t-1} + B^*_{t-1}}{A_{t-1}} - \overline{b} \right) + \epsilon^B_t$$

- Same attenuation of future tax forecasts:

$$\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}] = \theta \mathbb{E}_t[v^\tau_{t+1}]$$

$A_{t+1} = (1 + g_A)A_t$; G_t/A_t constant; taxes adjust

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- Match a smooth increase in debt/GDP over 50 years

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  - Initial BGP: debt/GDP $= 26\%$, NFA/GDP $= 4.3\%$

Calibration

Parameter	Value	Discipline
θ	0.092	Time series estimate
β	0.990	Calibrated quarterly MPC of 0.010
α	0.39	Average capital share
g_A	0.69%	Annual GDP growth of 2.78%
μ	3.94	Average rest-of-world/US GDP
ρ_B	0.978	Persistence of the tax present value
γ	1	Log utility
δ	2.2%	Annual depreciation of 8.6%

Quarterly parameters. Model MPC: 1%; estimated quarterly wealth MPC: 0.58%.

Government Borrowing and External Balances

$$\text{NFA}_t = -Q_t B_t^* \qquad \text{CA}_t = \text{NFA}_t - \text{NFA}_{t-1}$$

- Debt-financed transfers raise perceived household wealth

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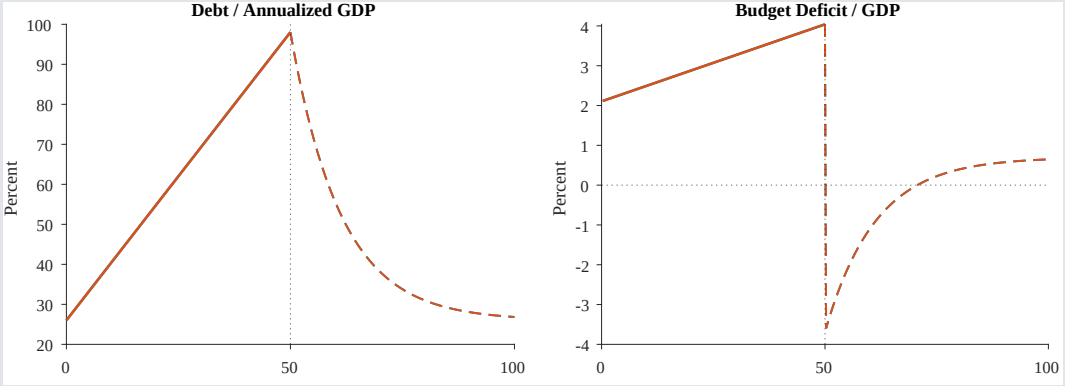
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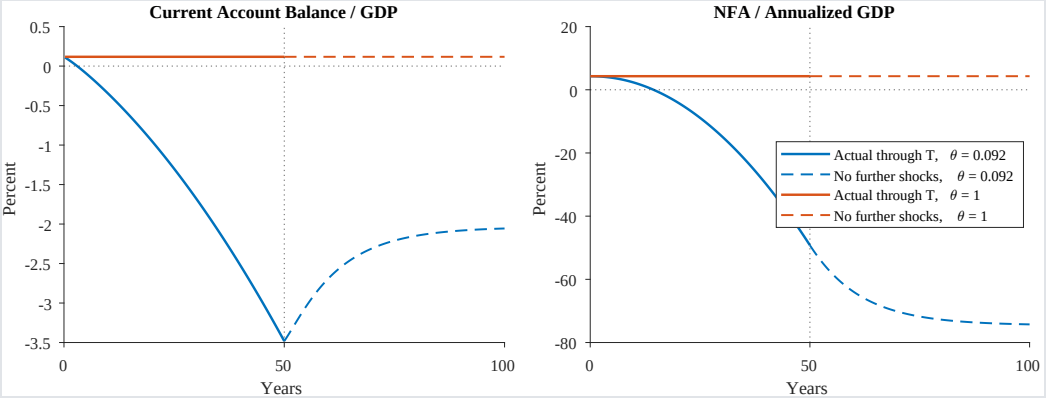
- Debt-financed transfers raise perceived household wealth
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- Current account deficits and declining NFA
- Foreign lending limits the crowding out of domestic capital

A Fifty-Year Increase in Government Debt: Paths



Blue: $\theta = 0.092$; orange: $\theta = 1$. Dashed: outcomes with no further shocks.

A Fifty-Year Increase in Government Debt: Paths



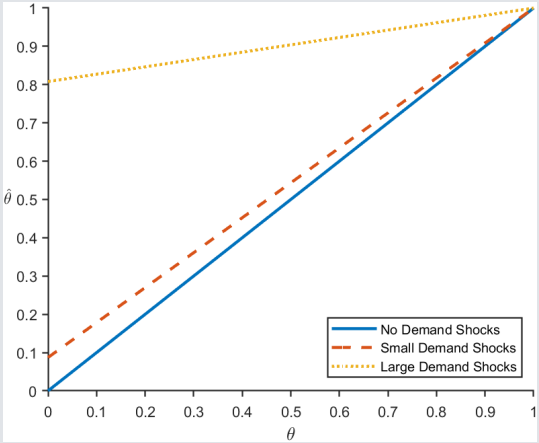
Blue: $\theta = 0.092$; orange: $\theta = 1$. Dashed: outcomes with no further shocks.

Fifty-Year Outcomes

	Data	$\theta = 0.092$	$\theta = 1$
Average budget deficit/GDP	3.78%	3.08%	3.08%
Average current account/GDP	−2.40%	−1.43%	0.12%
NFA/GDP in year 50	−71.1%	−49.2%	4.3%
Fifty-year I/Y trend	0.23%	−0.11%	0.00%

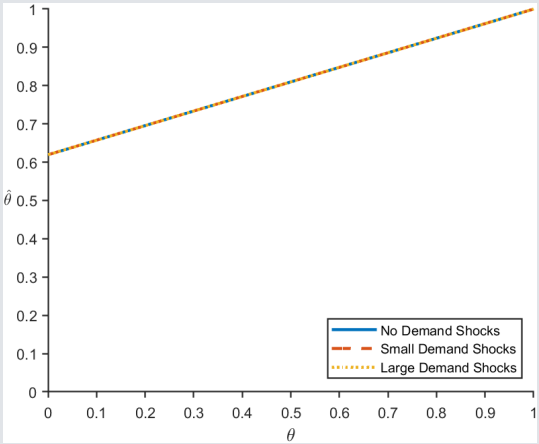
- Our non-Ricardian estimate explains the majority of the NFA decline and current account deficit
- Limited crowding out: I/Y falls 0.11 percentage points

Population Regression Exercise



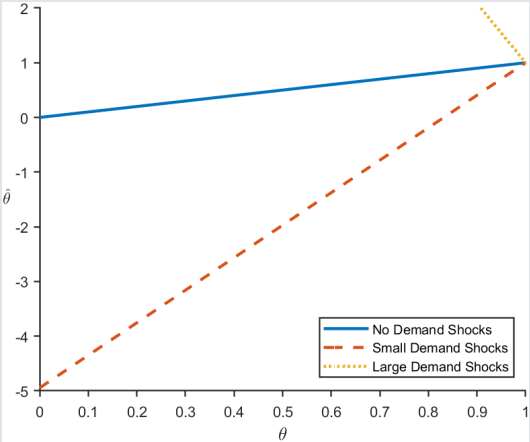
- OLS: biased with demand shocks

Population Regression Exercise



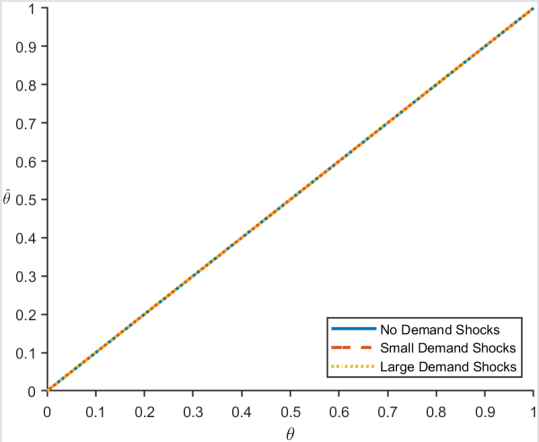
- OLS: biased with demand shocks
- Tax shocks alone: omit other consumption channels

Population Regression Exercise



- OLS: biased with demand shocks
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- Lagged IVs: invalid with persistent demand shocks

Population Regression Exercise



- OLS: biased with demand shocks
- Tax shocks alone: omit other consumption channels
- Lagged IVs: invalid with persistent demand shocks
- Macro shock IVs: recover θ

Conclusion

- How Ricardian Are We? *Not very.*
  - Households internalize 9.2% of future taxes
- Therefore, government borrowing has large external effects
  - Model explains majority of 50-year NFA decline and current account deficits
  - Open economy is crucial: limited crowding out of capital
- **B-HIVE**: many endogenous regressors + many macro shock IVs
  - Please reuse the library of structural shocks
  - jonathanjadams.com/structuralshocks

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